

Algorithmic Impact Assessment Results

Version: 0.10.0

Project Details

1. Name of Respondent

Hugo Parent

2. Job Title

Senior Program Advisor

3. Department

Employment and Social Development (Department of)

4. Branch

Integrity Services Branch

5. Project Title

Electronic Social Insurance Number Automation (eSINa)

6. Project ID from IT Plan

202400040

7. Departmental Program (from Department Results Framework)

Social Insurance Number Program

8. Project Phase

Implementation

[Points: +0]

9. Project Description

The electronic Social Insurance Number Automation (eSINa) initiative aims to increase the efficiency and effectiveness of electronic SIN (eSIN) service delivery. Automating standardized SIN processing is expected to create efficiencies by enabling eSIN clients to receive near real-time SIN services when certain criteria are met. This approach will help address the growing demand for SINs at Service Canada Centres (SCCs) and reduce Employment and Social Development Canada's (ESDC) / Service Canada's Centers dependency on paper-based processes.

The objective is to create a tool capable of determining the authenticity of identity documents uploaded by clients, as well as, reading, comparing, and extracting data from them. These documents will be submitted online as scans and/or photos (e.g.*.jpeg, *.pdf, etc.). Currently, there is no formal authentication process for secondary documents (e.g. provincial driver's licenses, passports, etc.), which are required for eSIN applications. Citizen Services Officers (CSOs) are responsible for reviewing and comparing the personal information against the Primary Identity Document (PID).

Clients who wish to obtain their SIN on the same day must visit a SCC in person. Alternatively, clients can submit a SIN application by mail, online and through the Newborn Registration Service (SIN@Birth). The service standards for these channels vary, ranging from 5 to 20 days. The proposed investment will explore opportunities to further automate eSIN processing by transitioning from manual processes to systematic services.

About the System

10. Please check which of the following capabilities apply to your system.

Image and object recognition: Analyzing very large data sets to automate the recognition, classification, and context associated with an image or object.,

Process optimization and workflow automation: Analyzing large data sets to identify and anomalies, cluster patterns, predict outcomes or ways to optimize; and automate specific workflows

[Points: +0]

Section 1: Impact Level: 1

Current Score: 31

Raw Score: 37

Mitigation Score: 37

Section 2: Requirements Specific to Impact Level 1

Peer review

None

Gender-based Analysis Plus

None

Notice

None

Human-in-the-loop-for-decisions

Decisions may be rendered without direct human involvement.

Explanation

In addition to any applicable legal requirement, ensure that a meaningful explanation is published for common decision results. The explanation must provide a general description of:

- the role of the system in the decision-making process;
- input data, its source and method of collection;
- the criteria used to evaluate input data and the operations applied to process it;
- the output produced by the system and any relevant information needed to interpret it in the context of the administrative decision; and
- the principal factors behind a decision.

Explanations must also inform clients of relevant recourse options, where appropriate.

Descriptions must be made available in plain language through the Algorithmic Impact Assessment and discoverable via a departmental website.

Training

None

IT and business continuity management

None

Approval for the system to operate

None

Other requirements

The Directive on Automated Decision-Making also includes other requirements that must be met for all impact levels.

tbs-sct.canada.ca/pol/doc-eng.aspx?id=32592

Contact your institution's ATIP office to discuss the requirement for a Privacy Impact Assessment as per the Directive on Privacy Impact Assessment.

Section 3: Questions and Answers

3.1: Impact Questions and Answers

3.1.1: Reasons for Automation

1. What is motivating your team to introduce automation into this decision-making process?

Lower transaction costs of an existing program, Use innovative approaches [Points: +0]

2. What client needs will the system address and how will this system meet them? If possible, describe how client needs have been identified.

By automating processing of eligible applications currently processed by agents, applications that currently take up to fifteen minutes to process will be completed in minutes. As a result, the system will address clients' need to receive their SIN online as soon as possible, enabling faster access to benefits and services.

3. Please describe any public benefits the system is expected to have.

Clients' applications will be processed more quickly, and they will have access to their SIN sooner. This will allow them to access benefits and services faster and without the need to visit a SCC in person.

4. How effective will the system likely be in meeting client needs?

Very effective [Points: +0]

5. Please describe any improvements, benefits, or advantages you expect from using an automated system. This could include relevant program indicators and performance targets.

Reduce the demand on in-person Service Canada Centres and decrease resources needed for manual eSIN processing.

Exceed program performance targets and severely reduce the processing times for eligible applications.

Reduce the reliance on paper and postage, which enhances environmental sustainability for the Department and brings down costs.

Enhance the accuracy of SIR data to strengthen the SIN Program's integrity and supports other dependent programs.

6. Please describe how you will ensure that the system is confined to addressing the client needs identified above.
The system will be operating based on predefined rules and will not be able to diverge from them.

7. Please describe any trade-offs between client interests and program objectives that you have considered during the design of the project.
No trade offs observed.

8. Have alternative non-automated processes been considered?
No [Points: +1]

9. What would be the consequence of not deploying the system?
Other (please specify) [Points: +0]

10. Other (please specify)
The potential time savings from automation-enhanced processing will not be realized. Service costs will remain stable rather than being reduced, and demand at SCCs will also remain stable.

3.1.2: Risk Profile

11. Is the project within an area of intense public scrutiny (e.g. because of privacy concerns) and/or frequent litigation?
No [Points: +0]

12. Are clients in this line of business particularly vulnerable?
No [Points: +0]

13. Are stakes of the decisions very high?
No [Points: +0]

14. Will this project have major impacts on staff, either in terms of their numbers or their roles?
No [Points: +0]

15. Will the use of the system create or exacerbate barriers for persons with disabilities?
No [Points: +0]

3.1.3: Project Authority

16. Will you require new policy authority for this project?
No [Points: +0]

3.1.4: About the Algorithm

17. The algorithm used will be a (trade) secret
No [Points: +0]

18. The algorithmic process will be difficult to interpret or to explain
No [Points: +0]

3.1.5: About the Decision

19. Please describe the decision(s) that will be automated.

Are the uploaded documents acceptable? Did they pass the authentication module?

Does the information on the primary identity document and secondary document match?

Are all the requirements met to process the SIN request systematically?

Can the SIN be issued?

If correspondence is required, which correspondence should be sent to the client via email?

20. Does the decision pertain to any of the categories below:

Social assistance (employment insurance, disability claims)

[Points: +1]

3.1.6: Impact Assessment

21. Which of the following best describes the type of automation you are planning?

Full automation (the system will make an administrative decision)

[Points: +4]

22. Please describe the role of the system in the decision-making process.

The role will be to initiate an automated decision-making process based on predefined rules.

23. Will the system be making decisions or assessments that require judgement or discretion?

No

[Points: +0]

24. Please describe the criteria used to evaluate client data and the operations applied to process it.

The criteria used is:

Accuracy, Authenticity, Consistency and Completeness

Operations:

The system will determine if a document provided is authentic (done by way of a confidence score)

If data on two different documents does not match the user will be advised

Once the user's application has been submitted it will be transferred eSIN automation program where it will be matched to fields and validated against the source.

25. Please describe the output produced by the system and any relevant information needed to interpret it in the context of the administrative decision.

The system will output :

A confidence score for each uploaded document - a tolerance threshold will be prescribed to determine if the document is acceptable or not.

Full text extraction via Optical Character Recognition (OCR) of each specified field on a document.

A confidence score for each section of extracted text - A tolerance threshold will be prescribed to determine if it's acceptable or not.

An extract of all information submitted by the client in their eSIN application - this

information will be used to populate fields in eSIN automation program. When a decision point will be reached based on this information, the system will be following the programmed rules.

A decision on whether the application will be processed systematically or not will be sent by the eSIN automation program to MSD to update the status of the application.

26. Will the system perform an assessment or other operation that would not otherwise be completed by a human?

No

[Points: +0]

27. Is the system used by a different part of the organization than the ones who developed it?

No

[Points: +0]

28. Are the impacts resulting from the decision reversible?

Reversible

[Points: +1]

29. How long will impacts from the decision last?

Impacts are most likely to be brief

[Points: +1]

30. Please describe why the impacts resulting from the decision are as per selected option above.

If, in the course of processing, a SIN eligibility decision is made by the system it is considered final.

If, in the course of processing, a request is rejected to manual processing, a SIN eligibility decision will be made by an agent and will be considered final.

Whether the decision is made by the system or by an individual, a reassessment will take place if one is requested (it will be processed manually), or if a new request is received (it may be processed manually or electronically, depending on how it is submitted).

31. The impacts that the decision will have on the rights or freedoms of individuals will likely be:

Little to no impact

[Points: +1]

32. Please describe why the impacts resulting from the decision are as per selected option above.

As the SIN application process has no impact on the rights or freedoms of individuals, there are no impacts from the decision.

33. The impacts that the decision will have on the equality, dignity, privacy, and autonomy of individuals will likely be:

Little to no impact

[Points: +1]

34. Please describe why the impacts resulting from the decision are as per selected option above.

As the SIN application process has no impact on the equality, dignity, privacy, and autonomy of individuals, there are no impacts from the decision.

35. The impacts that the decision will have on the health and well-being of individuals will likely be:

Little to no impact

[Points: +1]

36. Please describe why the impacts resulting from the decision are as per selected option above.

As the SIN application process has no impact on the health and well-being of individuals, there are no impacts from the decision.

37. The impacts that the decision will have on the economic interests of individuals will likely be:

Little to no impact

[Points: +1]

38. Please describe why the impacts resulting from the decision are as per selected option above.

This decision will not have any impact on the economic interests of individuals that is any different from the current way of processing SIN applications.

39. The impacts that the decision will have on the ongoing sustainability of an environmental ecosystem, will likely be:

Little to no impact

[Points: +1]

40. Please describe why the impacts resulting from the decision are as per selected option above.

This decision will not have any impact on the ongoing sustainability of an environmental ecosystem that is any different from the current way of processing SIN applications.

3.1.7: About the Data - Data Source

41. Will the Automated Decision System use personal information as input data?

Yes

[Points: +4]

42. Have you verified that the use of personal information is limited to only what is directly related to delivering a program or service?

Yes

[Points: +0]

43. Is the personal information of individuals being used in a decision-making process that directly affects those individuals?

Yes

[Points: +2]

44. Have you verified if the system is using personal information in a way that is consistent with: (a) the current Personal Information Banks (PIBs) and Privacy Impact Assessments (PIAs) of your programs or (b) planned or implemented modifications to the PIBs or PIAs that take new uses and processes into account?

Yes

[Points: +0]

45. Please list relevant PIB Bank Numbers.

PSU 905, PSU 933, PSU 913, PSU 942,

46. What is the highest security classification of the input data used by the system? (Select one)

Protected B / Protected C

[Points: +3]

47. Who controls the data?

Federal government

[Points: +1]

48. Will the system use data from multiple different sources?

Yes

[Points: +4]

49. Will the system require input data from an Internet- or telephony-connected device? (e.g. Internet of Things, sensor)

No

[Points: +0]

50. Will the system interface with other IT systems?

Yes

[Points: +4]

51. Who collected the data used for training the system?

Your institution

[Points: +1]

52. Who collected the input data used by the system?

Your institution

[Points: +1]

53. Please describe the input data collected and used by the system, its source, and method of collection.

Input data collected: information found on PID and secondary documents (government issued identity documents)

Source: Clients or someone applying on their behalf who provided.

Method of collection: These documents were uploaded at application.

3.1.8: About the Data - Type of Data

54. Will the system require the analysis of unstructured data to render a recommendation or a decision?

Yes

[Points: +0]

55. What types of unstructured data? (Check all that apply)

Images and videos

[Points: +4]

3.2: Mitigation Questions and Answers

3.2.1: Consultations - Have you engaged with any of the following groups?

1. Internal Stakeholders (federal institutions, including the federal public service)

Yes

[Points: +1]

2. Which Internal Stakeholders have you engaged?

Access to Information and Privacy Office, Program Policy, Human Resources, Communications services, Accessibility Working Groups, Data Governance

[Points: +0]

3. External Stakeholders (groups in other sectors or jurisdictions)

No

[Points: +0]

3.2.2: De-Risking and Mitigation Measures - Data Quality

4. Do you have documented processes in place to test datasets against biases and other unexpected outcomes? This could include experience in applying frameworks, methods, guidelines or other assessment tools.

Yes

[Points: +2]

5. Is this information publicly available?

No

[Points: +0]

6. Have you developed a process to document how data quality issues were resolved during the design process?

Yes

[Points: +1]

7. Is this information publicly available?

No

[Points: +0]

8. Have you undertaken a Gender Based Analysis Plus of the data?

Yes

[Points: +1]

9. Is this information publicly available?
No [Points: +0]

10. Have you assigned accountability in your institution for the design, development, maintenance, and improvement of the system?
Yes [Points: +2]

11. Do you have a documented process to manage the risk that outdated or unreliable data is used to make an automated decision?
Yes [Points: +2]

12. Is this information publicly available?
No [Points: +0]

13. Is the data used for this system posted on the Open Government Portal?
No [Points: +0]

3.2.3: De-Risking and Mitigation Measures - Procedural Fairness

14. Does the audit trail identify the authority or delegated authority identified in legislation?
Yes [Points: +1]

15. Does the system provide an audit trail that records all the recommendations or decisions made by the system?
Yes [Points: +2]

16. Are all key decision points identifiable in the audit trail?
Yes [Points: +2]

17. Are all key decision points within the system's logic linked to the relevant legislation, policy or procedure?
Yes [Points: +1]

18. Do you maintain a log detailing all of the changes made to the model and the system?
Yes [Points: +2]

19. Does the system's audit trail clearly set out all decision points made by the system?
Yes [Points: +1]

20. Can the audit trail generated by the system be used to help generate a notification of the decision (including a statement of reasons or other notification) where required?
Yes [Points: +1]

21. Does the audit trail identify precisely which version of the system was used for each decision it supports?
Yes [Points: +2]

22. Does the audit trail show who the authorized decision-maker is?
Yes [Points: +1]

23. Is the system able to produce reasons for its decisions or recommendations when required?
Yes [Points: +2]

24. Is there a process in place to grant, monitor, and revoke access permission to the system?
Yes [Points: +1]

25. Is there a mechanism to capture feedback by users of the system?
Yes [Points: +1]

26. Is there a recourse process planned or established for clients that wish to challenge the decision?

Yes

[Points: +2]

27. Does the system enable human override of system decisions?

Yes

[Points: +2]

28. Is there a process in place to log the instances when overrides were performed?

Yes

[Points: +1]

29. Does the system's audit trail include change control processes to record modifications to the system's operation or performance?

Yes

[Points: +2]

30. Have you prepared a concept case to the Government of Canada Enterprise Architecture Review Board?

No

[Points: +0]
- 3.2.4: De-Risking and Mitigation Measures - Privacy
31. If your system uses or creates personal information, have you undertaken or have you undertaken a Privacy Impact Assessment, or updated an existing one?

Yes

[Points: +1]

32. Please indicate the following in your answer: Title and scope of the Privacy Impact Assessment; How the automation project fits into the program; and Date of Privacy Impact Assessment completion or modification.

Privacy Impact Assessment Report for Electronic Social Insurance Number Automation (eSINa)

33. Have you design and build security and privacy into your systems from the concept stage of the project?

Yes

[Points: +1]

34. Is the information used within a closed system (i.e. no connections to the Internet, Intranet or any other system)?

No

[Points: +0]

35. If the sharing of personal information is involved, has an agreement or arrangement with appropriate safeguards been established?

Yes

[Points: +1]

36. Will you de-identify any personal information used or created by the system at any point in the lifecycle?

Yes

[Points: +1]

37. Please describe your de-identification method(s).

eSINA's automated decision system for personal identification documents verification does not retain any personal information and as such, de-identification is not required. The original scanned documents and the extract of all information they contain will be deleted after a predetermined period.